	Autoware built-in diagnostics	Won 2023 LiDAR bitmap	Zhang 2024 LiDAR distribution	Lee 2022 camera CNN	Yu 2021 state χ^2 test	Huang & Su EKF residual	RADAS stream rate	Alibi Detect online MMD (nf-calibrated)	Proposed
LiDAR sparsification (mid)	0/5	0/5	5/5 0.8 s	0/5				0/5	0/5
LiDAR sparsification (high)	0/5	1/5 130.4 s	5/5 0.9 s	0/5			0/1	5/5 5.4 s	5/5 1.2 s
LiDAR clutter (mid)	0/5	0/5	5/5 0.7 s	0/5				0/5	5/5 1.2 s
LiDAR clutter (high)	0/5	0/5	5/5 0.6 s	0/5				0/5	5/5 1.2 s
LiDAR death	45/45 4.0 s				0/45	0/45	1/1 3.8 s		45/45 1.4 s
Camera noise (mid)	0/5	0/5	0/5	5/5 0.7 s				5/5 20.6 s	5/5 1.3 s
Camera noise (high)	0/5	0/5	0/5	5/5 0.9 s				5/5 17.4 s	5/5 1.3 s
Camera death	0/45				0/45	0/45	1/1 2.4 s		45/45 1.4 s
GNSS position noise (mid)	0/45				45/45 1.0 s	45/45 1.1 s		0/6	45/45 1.9 s
GNSS position noise (high)	0/45				45/45 1.0 s	45/45 1.0 s	0/1	0/5	45/45 1.2 s
GNSS correlated bias (mid)	0/45				45/45 1.2 s	45/45 1.3 s		0/5	45/45 2.0 s
GNSS correlated bias (high)	0/45				45/45 1.1 s	45/45 1.1 s		0/5	45/45 1.5 s
GNSS outage	0/45				0/45	0/45		0/5	45/45 2.2 s
IMU gyro noise (mid)	45/45 38.6 s				45/45 3.3 s	45/45 1.0 s	0/1	0/5	45/45 4.7 s
IMU gyro noise (high)	45/45 6.9 s				45/45 1.5 s	45/45 1.0 s		0/5	45/45 3.1 s
IMU gyro bias (mid)	45/45 6.2 s				45/45 1.5 s	45/45 1.5 s		5/5 3.7 s	45/45 3.2 s
IMU gyro bias (high)	45/45 5.6 s				45/45 1.2 s	45/45 1.0 s		5/5 6.4 s	45/45 3.1 s
IMU bias ramp (mid)	45/45 15.7 s				45/45 14.1 s	45/45 8.4 s		0/5	45/45 10.0 s
IMU bias ramp (high)	45/45 10.6 s				45/45 5.8 s	45/45 4.5 s		0/6	45/45 5.9 s

Cell: detections / runs (median latency, simulation s). Hatched: detector does not consume that sensor or was not run.
Navigation rows and camera/LiDAR death: 45 runs per cell (bag replay); LiDAR/camera degradation rows: 5 runs (raw 2 Hz recordings); Alibi MMD: 5–6; RADAS: 1.
Proposed lso-mid 0/5 is by design (no capability loss, no intervention).
False alarms — navigation evaluation (26 no-fault runs + 597 pre-injection windows): Proposed 1, Autoware diag. 1 (same run), Yu 0, Huang & Su 0;
LiDAR/camera evaluation (3 no-fault runs + 30 windows): 0 for all; Alibi (nf-calibrated): 3 of 8 no-fault run×sensor evaluations; RADAS: 0 of 5 pre-injection windows.